

Discrete Probability Distributions and Applications

MGMT 670: Business Analytics



Topics

- Random variable
 - Expected value
 - Variance, Stdev
- Combination of random variables
 - Covariance
 - Correlation
- Portfolio management
- Binomial Random variable
 - Mean
 - Standard deviation

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Random Variable

- Sure Thing: pays \$5
- Gamble:
 - pays \$10 with probability 0.4
 - pays \$0 with probability 0.6

Payoff (X) from gamble is a random variable.

It's distribution is:

$$P(X = 10) = 0.4$$

$$P(X = 0) = 0.6$$

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Random Variable

- **Random variable: assigns a probability value to numerical outcomes!**
 - Discrete random variable
 - Countable number of outcomes
 - Continuous random variable
 - Infinite, uncountable number of outcomes

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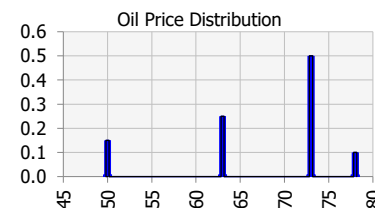
Example: Oil Commodities



- A commodities investor is concerned with the price of oil for the coming year.
- The investor believes there are four possible scenarios for the oil market in the coming year: high demand, moderate demand, no growth, or moderate contraction.
- She estimates that the price of oil per barrel in each case will be \$78, 73, 63, and 50, respectively.
- Also, she has assessed that the probabilities of these outcomes are 0.10, 0.50, 0.25, and 0.15, respectively.

Distribution of Oil Price (X)

Economic Outcome	Price, x_i	Probability
High Demand	78	0.10
Moderate Demand	73	0.50
No Growth	63	0.25
Moderate Contraction	50	0.15
Total		1.00



Expected Value of Discrete Random Variable

- Weighted average of all possible values

$$E(X) = \mu = \sum_i x_i f(x_i)$$

x_i	$f(x_i)$	$x_i f(x_i)$	
78	0.10	7.80	
73	0.50	36.50	
63	0.25	15.75	
50	0.15	7.50	
Sum	1.00	67.55	$= E(X)$

Variance of Discrete Random Variable

- Weighted average of squared deviations from the mean

$$\text{Var}(X) = \sigma^2 = E[(X - E(X))^2] = \sum_i (x_i - E(X))^2 f(x_i)$$

x_i	$f(x_i)$	$x_i f(x_i)$	$(x_i - \mu)^2$	$(x_i - \mu)^2 f(x_i)$	
78	0.10	7.80	109.2025	10.920	
73	0.50	36.50	29.7025	14.851	
63	0.25	15.75	20.7025	5.176	
50	0.15	7.50	308.0025	46.200	
Sum	1.00	67.55		77.148	$= \sigma^2$

Standard Deviation of Discrete Random Variable

- Weighted average of deviations from the mean

$$\text{stdev}(X) = \sigma = \sqrt{\sigma^2} = \sqrt{\sum_i (x_i - E(X))^2 f(x_i)}$$

$$\sqrt{\sigma^2} = \sqrt{77.148} = 8.783$$

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Example: Powerball

NUMBERS MATCHED	PRIZE	PRIZE - COST	LIKELIHOOD	PROBABILITY	(PRIZE - COST) X PROBABILITY
5 white and red	\$275,400,000	\$275,399,998	1 in 292,201,338	0.0000000034	\$0.94
5 white	\$1,000,000	\$999,998	1 in 11,688,053.52	0.0000000856	\$0.09
4 white and red	\$50,000	\$49,998	1 in 913,129.18	0.0000010951	\$0.05
4 white	\$100	\$98	1 in 36,525.17	0.0000273784	\$0.00
3 white and red	\$100	\$98	1 in 14,494.11	0.0000689935	\$0.01
3 white	\$7	\$5	1 in 579.76	0.0017248517	\$0.01
2 white and red	\$7	\$5	1 in 701.33	0.0014258623	\$0.01
1 white and red	\$4	\$2	1 in 91.98	0.0108719287	\$0.02
Red	\$4	\$2	1 in 38.32	0.0260960334	\$0.05
Nothing	\$0	-\$2	1 in 1.04	0.9597837679	-\$1.92
					EXPECTED VALUE: -\$0.74

SOURCE: Business Insider calculations with odds from powerball.com

Jackpot: 275.4 million!

EXAMPLES, PRACTICE QUESTIONS

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Example: Powerball



Match	Prize	Odds
	Grand Prize	1 in 292,201,338.00
	\$1,000,000	1 in 11,688,053.52
	\$50,000	1 in 913,129.18
	\$100	1 in 36,525.17
	\$100	1 in 14,494.11
	\$7	1 in 579.76
	\$7	1 in 701.33
	\$4	1 in 91.98
	\$4	1 in 38.32

What's the expected return?

<https://www.youtube.com/watch?v=SEtSiUfpymA>

EXAMPLES, PRACTICE QUESTIONS

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Analyzing Population Parameters

- The mean or expected return **does not imply**
 - that the most likely oil price is 67.55
 - 67.55 is not a possible value
- If we **repeat the experiment infinite many times**, the prices of oil will be different each time
 - Expected value will be the average value
 - Standard deviation measures variation in prices

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Examples

- X is the random variable for the return rate of EXXON stock (in percentage)

Economic Outcome	Rate of return, x_i	Probability
High Demand	45%	0.6
Moderate Demand	10%	0.1
Contraction	-25%	0.3
Total		1.00

Compute

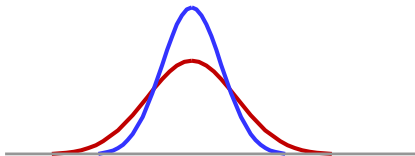
- $E(X)=?$
- $\text{stdev}(X)=?$
- What are the interpretations of these numbers?

Y is the random variable for the return rate of Apple stock (in percentage). $Y = -40$ with probability 0.2; $Y = 70$ with probability 0.8.

- $E(Y)=?$
- $\text{stdev}(Y)=?$
- What are the interpretations of these number?

Comparing two distributions

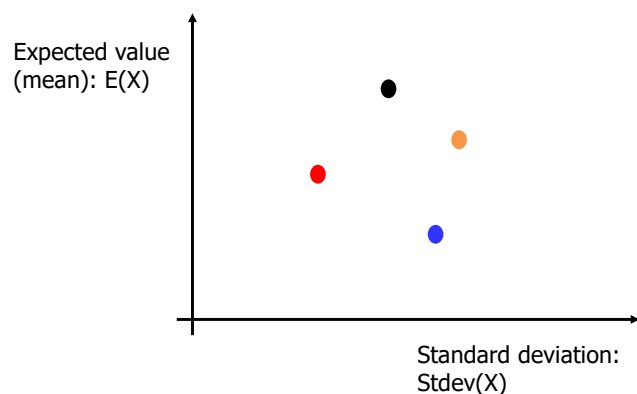
- Among two distributions of the same mean, the one with higher standard deviation is more “risky”, and usually less preferred.



Reducing risks!

- Reducing standard deviation is important in investment and management.
- Diversification can substantially reduce risks.
- Examples
 - Investment: invest in different types of stocks
 - Supply chains: getting supplies from multiple, independent suppliers
 - **Other examples?**

Which dots do you like?



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Rules for Expectations & Variance

Assume X, Y are random variables, and a, b are constants.

$$E(aX + b) = aE(X) + b$$

$$E(aX \pm bY) = aE(X) \pm bE(Y)$$

$$\text{Var}(aX + bY) = a^2\text{Var}(X) + b^2\text{Var}(Y) + 2ab\text{Cov}(X, Y)$$

Tattoo this somewhere
on your body

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How Diversification Reduces Risk

Suppose the proportions of money invested in the first stock is p ; second stock is $1-p$

- Expected Return

$$= pE(X) + (1-p)E(Y)$$
- Variance of Portfolio

$$= \text{Var}(pX + (1-p)Y)$$

$$= p^2 \sigma_x^2 + (1-p)^2 \sigma_y^2 + 2p(1-p) \text{Cov}(X, Y)$$

$$= p^2 \sigma_x^2 + (1-p)^2 \sigma_y^2 + 2p(1-p) \rho_{xy} \sigma_x \sigma_y$$

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Investment Diversification

Now consider investing 40% of the total money on EXXON and 60% on APPLE.

$$Z = 0.4X + 0.6Y$$

- What's the expected return of this portfolio? That is if you invest 1 dollar in this portfolio, how much are you getting in expectation next year?
 $E(Z) = ?$
- How about the risk of this portfolio if the correlation of X, Y is -0.111 ?
 $\text{Var}(Z) = ?$ $\text{Stdev}(Z) = ?$

EXAMPLES, PRACTICE QUESTIONS

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$$Var(aX + bY) = a^2Var(X) + b^2Var(Y) + 2abCov(X, Y).$$

How Diversification Reduces Risk

$$E(X) = 20.5, \quad Var(X) = 31.5^2,$$

$$E(Y) = 48, \quad Var(Y) = 44^2, \quad \rho_{xy} = 0.05$$

$$Z = pX + (1-p)Y$$

p	E(Z)	var(Z)	stdev(Z)
1	20.5	992.25	31.5
0.95	21.875	906.9291	30.11526
0.9	23.25	835.5565	28.90599
0.85	24.625	778.1321	27.89502
0.8	26	734.656	27.10454
0.75	27.375	705.1281	26.55425
0.7	28.75	689.5485	26.25926
0.65	30.125	687.9171	26.22817
0.6	31.5	700.234	26.46193
0.55	32.875	726.4991	26.95365
0.5	34.25	766.7125	27.68957
0.45	35.625	820.8741	28.6509
0.4	37	888.984	29.81583
0.35	38.375	971.0421	31.16155
0.3	39.75	1067.049	32.66571
0.25	41.125	1177.003	34.30748
0.2	42.5	1300.906	36.06807
0.15	43.875	1438.757	37.93095
0.1	45.25	1590.557	39.88178
0.05	46.625	1756.304	41.90828
0	48	1936	44

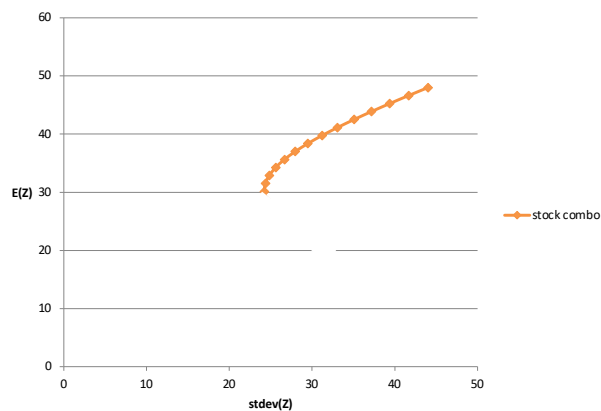
Greater Diversification, Lower Risk

By diversifying a portfolio with international stocks, real estate, and even commodities and hedge funds, you can lower your risk. That makes it easier to plan for retirement and other financial goals. Here, we look at the range of likely outcomes for two \$10 million portfolios. One is made up of just U.S. stocks and bonds, while the other adds international stocks and bonds, as well as some alternative investments.

HORIZON	RANGE OF OUTCOMES*	U.S. STOCKS, BONDS & BONDS	U.S. STOCKS, BONDS INT'L & ALTERNATIVE INVESTMENTS
10 Years	High	\$26.8 million	\$24.1 million
	Low	10.6	12.6
20 Years	High	52.7	47.3
	Low	15.0	18.5

*90% of the expected outcomes fall within this range
Data: JPMorgan Private Bank

Efficient Frontier



Binomial Distributions

- The experiment consists of a sequence of *n identical, independent* trials.
- Two outcomes, *success* and *failure*, are possible on each trial.
- The probability of a success, denoted by *p*, does not change from trial to trial.
- Binomial(*n*,*p*): number of *successes in this experiment*.

Example



- Alice
 - 6 calls per day ; success rate 40%
- Bob
 - 8 calls per day success rate 30%
- Which one would you hire?

Example



- Alice
 - 6 calls per day; success rate 40%
 Binomial(6,0.4)
- Bob
 - 8 calls per day; success rate 30%
 Binomial(8,0.3)

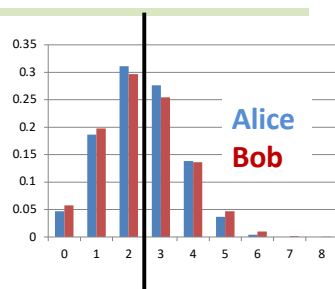
Mean and standard deviation

Alice: 6 calls per day ; success rate 40%
 Bob : 8 calls per day; success rate 30%
 On average how many sales per day?

For binomial distributions: mean = np

For binomial distribution :

standard dev = $\sqrt{np(1-p)}$



$$6 \times .4 = 8 \times .3 = 2.4$$

$$Stdev_{Alice} = \sqrt{6 \times .4 \times (1-.4)} = 1.2$$

$$Stdev_{Bob} = \sqrt{8 \times .3 \times (1-.3)} = 1.296$$

Alice: Binomial(6;.4)

$$f(0) = P(\text{Alice having 0 sale})=?$$

$$f(1) = P(\text{Alice having 1 sale})=?$$

$$f(2) = P(\text{Alice having 2 sales})=?$$

$$f(3) = P(\text{Alice having 3 sales})=?$$

$$f(4) = P(\text{Alice having 4 sales})=?$$

$$f(5) = P(\text{Alice having 5 sales})=?$$

$$f(6) = P(\text{Alice having 6 sales})=?$$

$P(0 \text{ sale in 6 calls}) =$

$$1(1-.4)^6 = 0.0466$$

scenarios in which no call is successful

probability that all calls are NOT successful

$P(1 \text{ sale in 6 calls}) =$

$$6(.4)^1(1-.4)^5 = 0.186624$$

of scenarios in which exactly 1 call is successful

probability that 1 specific call is successful, and the remaining 5 calls are not.

$P(2 \text{ sales in 6 calls}) =$

$$15(.4)^2(1-.4)^4 = 0.31104$$

of scenarios in which exactly two calls are successful

probability of 2 specific calls are successful and the remaining 4 calls are not.

The Binomial Probability Distribution

- Binomial Probability Function

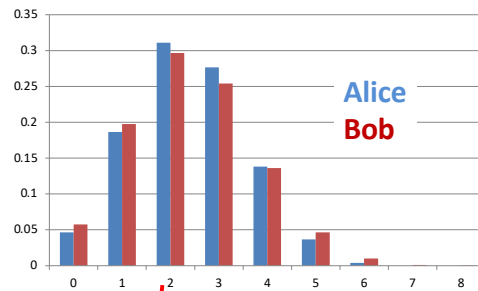
$$f(x) = \frac{n!}{x!(n-x)!} p^x (1-p)^{(n-x)}$$

where

- $f(x)$ = the probability of x successes in n trials
 - n = the number of trials
 - p = the probability of success on any one trial
- $n!$ means $n(n-1)(n-2)\dots(2)(1)$

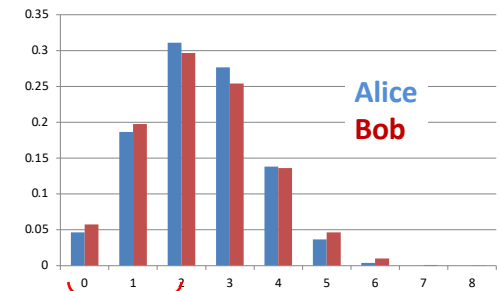
EXCEL: BINOMDIST(#success, #trials, p_success, 0)

	Alice	Bob
0	0.046656	0.057648
1	0.186624	0.19765
2	0.31104	0.296475
3	0.27648	0.254122
4	0.13824	0.136137
5	0.036864	0.046675
6	0.004096	0.010002
7	0	0.001225
8	0	6.56E-05



EXCEL:
BINOMDIST(2, #trials, p_success, 0)

	Alice	Bob
0	0.046656	0.057648
1	0.186624	0.19765
2	0.31104	0.296475
3	0.27648	0.254122
4	0.13824	0.136137
5	0.036864	0.046675
6	0.004096	0.010002
7	0	0.001225
8	0	6.56E-05



EXCEL:
BINOMDIST(2, #trials, p_success, 1)

To evaluate **how bad** the sale-person is:

P(Making 0 sale)=?

P(Making at most 1 sale)=?

P(Making at most 2 sales)=?

....

Illustration

- <https://istats.shinyapps.io/BinomialDist/>

Example 2

In 2024 president election, Trump gets 58.6% of votes in IN. In a random collection of 10 voters.

- On average, how many votes does Trump have?
- What is its standard deviation?
- What's the probability that less than 9 of these randomly selected 10 voters voted Trump

Example 3

The records of a department store show that 20% of its customers who make a purchase return the merchandise in order to exchange it. This week the store had 150 customers. Use Excel to compute:

- what is the probability that less than 20 customers will return the merchandise for exchange?
`=BINOM.DIST(19, 150, 0.2, TRUE)`
- what is the probability that more than 40 customers will return the merchandise for exchange?
- what is the probability that between 25 and 35 customers will return the merchandise for exchange?

Case Study (Airplane Booking)

- An airplane has 100 seats. The airline “overbooks” a flight (sells more tickets than available seats) in order to maximize their profit. Assume that each ticket holder’s decision to show up for a flight is an independent Bernoulli trial with a probability of showing up for the flight of 0.92. If the airline profit is \$10 for each seat sold and the airline loses \$40 for each “bumped” passenger, what is the expected profit if 103 seats are sold?

Summary

- Random variables: continuous and discrete
- Discrete probability distribution
 - Conditions
 - Expected value, variance, and covariance
- Rules for expectations and variances
- Portfolio Management
- Binomial distribution
 - Evaluation of the probability of an event
 - Mean, variance, and standard deviation